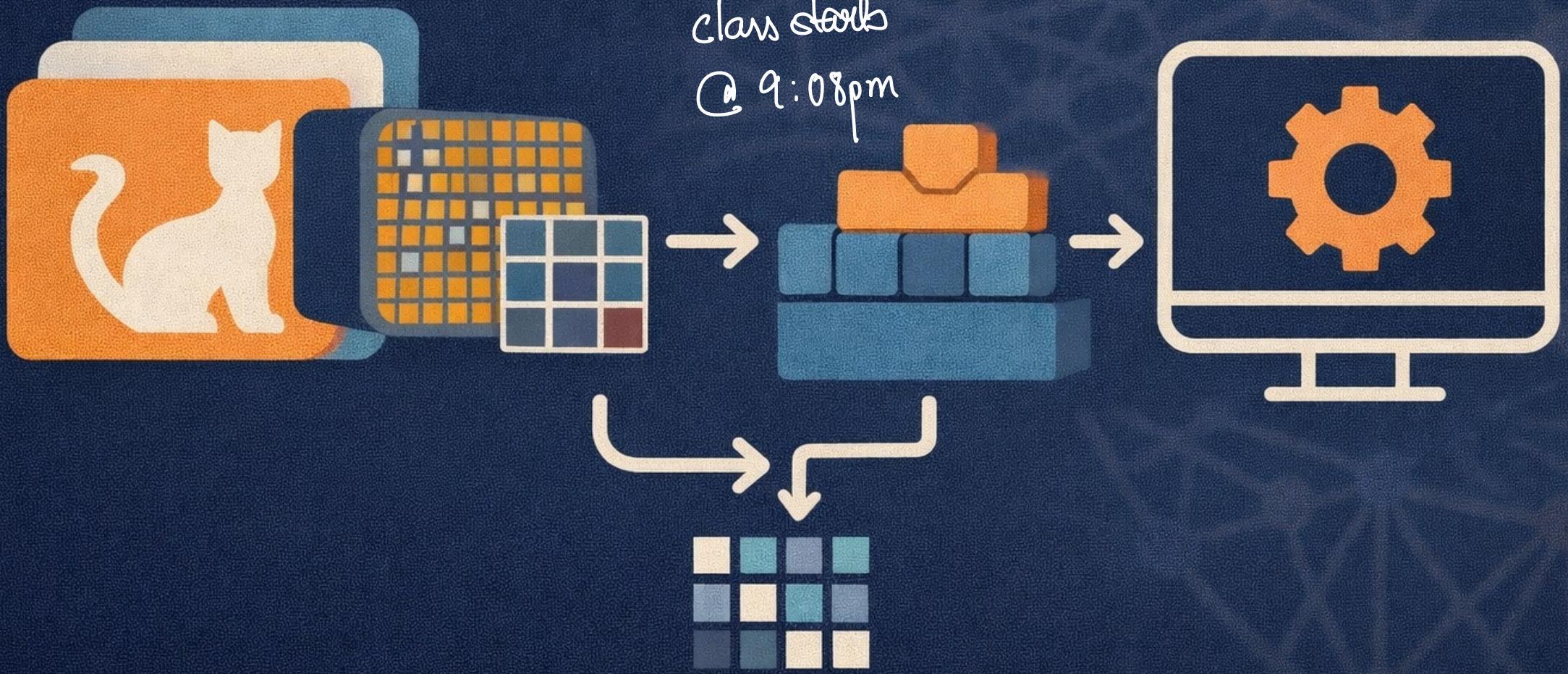
CONVOLUTIONAL NEURAL NETWORK



Lecture : Strides, Pooling and Overfitting

#Agenda

1) Recap (Idea of Convolution)
2) Stride
3) pooling
4) Overfitting.

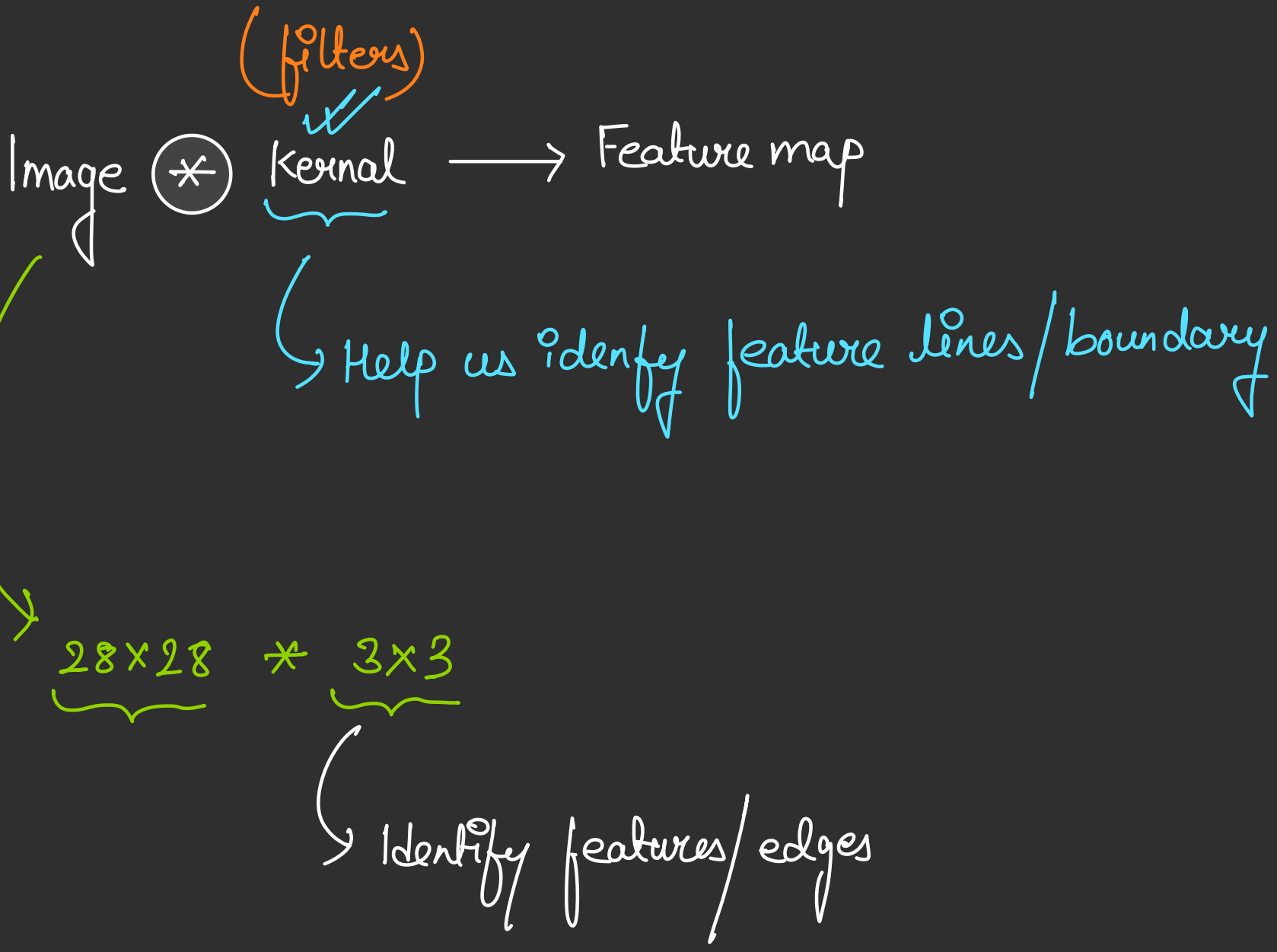
Recap

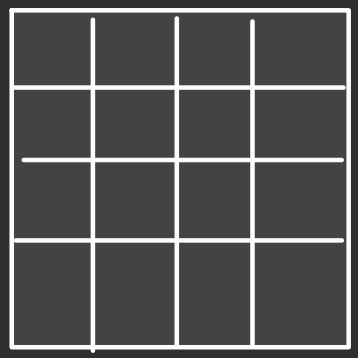
Image classification ⟶ ANN

28×28 ⟶ flatten (784) / feature ⟶ NN

28×28×3 (784 × 3)

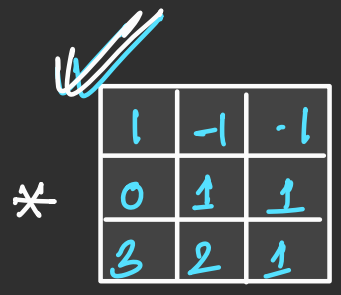
RGB


(filters)
Image * Kernal → Feature map
↳ Help us identify feature lines / boundary
28×28 * 3×3
↳ Identify features / edges



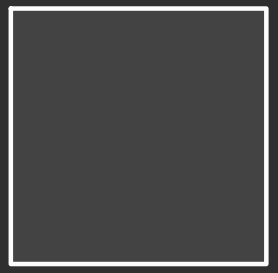
Image

(4x4)



Kernal

(3x3)



Feature map

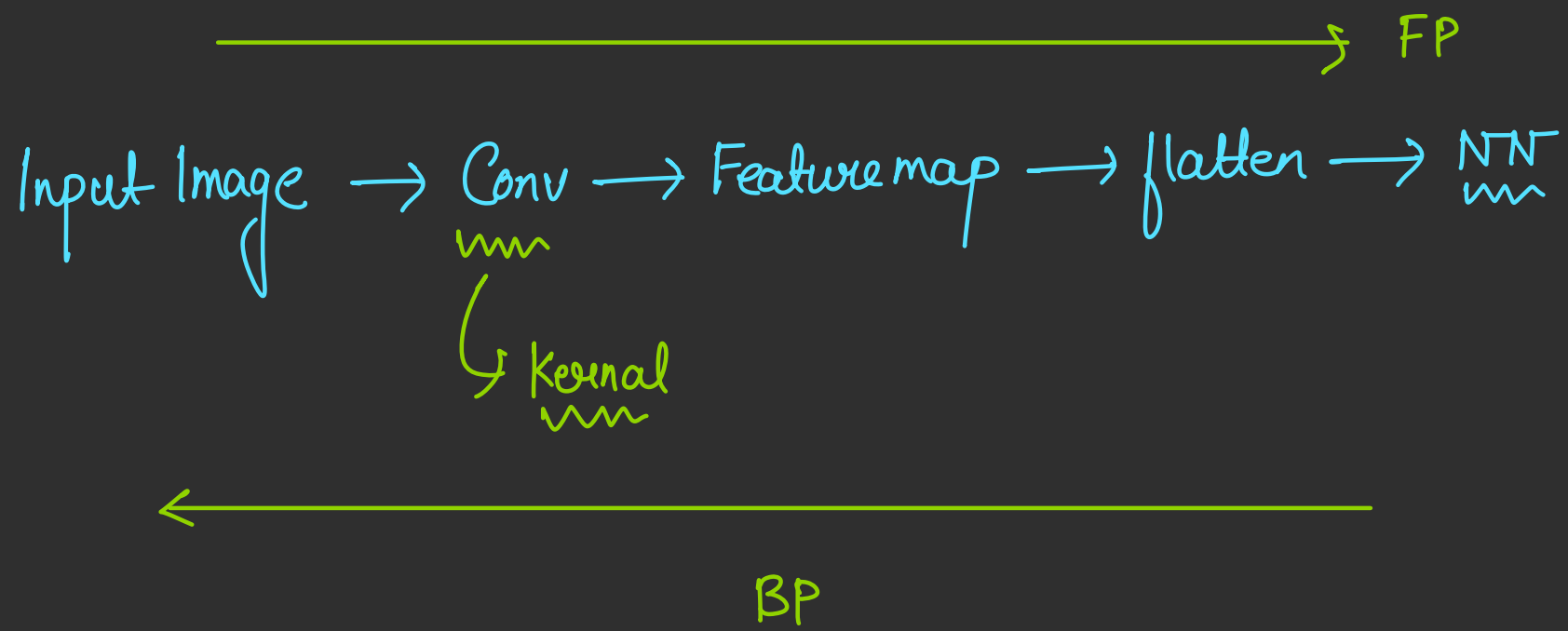
weights

RGB

4x4x3

Kernal

(3x3x3)


FP
Input Image
Conv
Feature map
flatten
NN
Kernal
BP

Output feature map size $= \left(\frac{n+2p-f}{s} + 1\right)$

- $n \Rightarrow$ size of input image
- $p \Rightarrow$ padding
- $s \Rightarrow$ stride
- $f \Rightarrow$ filter size

$\begin{cases} s = 1 \\ 2p \Rightarrow 0 \end{cases}$

Input = 5x5

filter $\Rightarrow$ 3x3

$\frac{5+0-3}{1} \Rightarrow 2+1 = 3$

3x3

output / feature map

Strides

np.arange(1, 10, 2)



$$\left(\frac{n + 2p - f}{s} + 1\right)$$

$s = 1$ $\Rightarrow$ $\left(\frac{5+0-3}{1} + 1\right) \Rightarrow 3 \times 3$

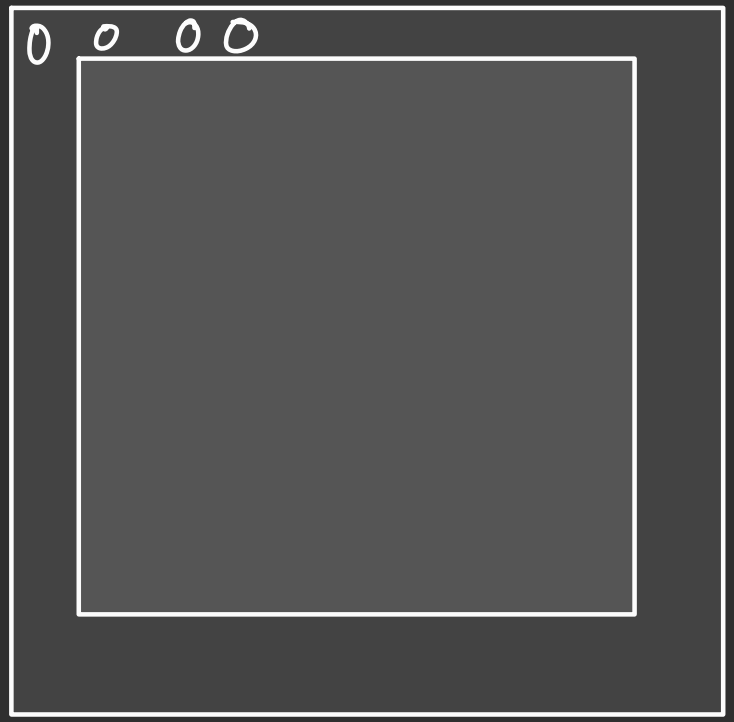
$s = 2$ $\left(\frac{5+0-3}{2} + 1\right) \Rightarrow 2 \times 2$

padding

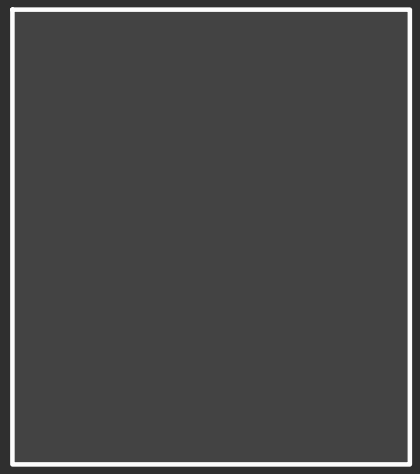
zero padding

same

valid



same Vs Valid



I/P

(28x28)

(3x3)

28x28

Feature map

26x26

Feature map

How to tackle an overfitting problem in CNN?

1) Modify your architecture / model pipeline.

2) Modify the data

"Data augmentation"

Summary

Input Image → ConvD → Feature map → Flatten → NN

Stride - You skip some information but helps to train faster

Skipping pixel in images

padding → same (Padding zero)

valid (No padding)

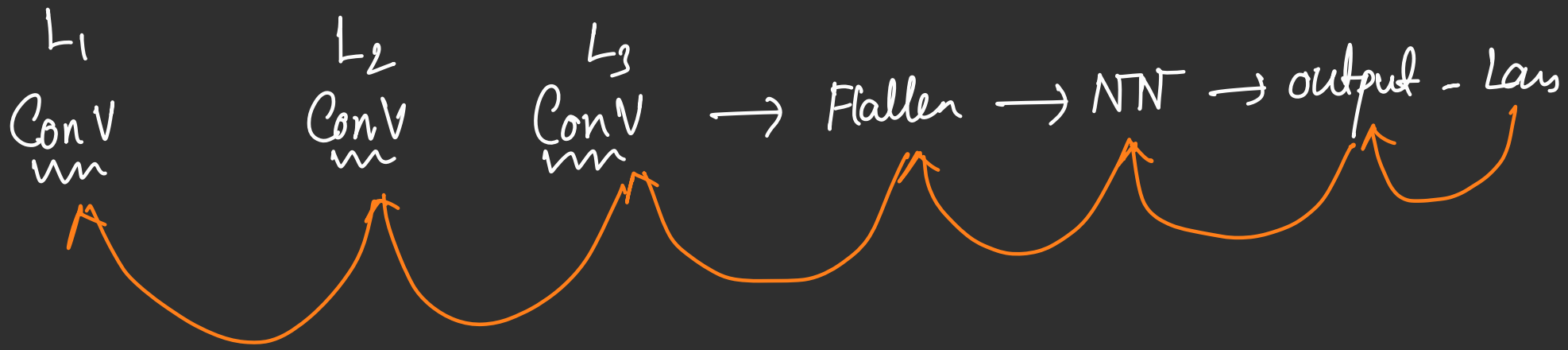
$$\text{Feature map size} = \left(\frac{n + 2p - f}{s} + 1 \right)$$

pooling ⟶ To reduce feature map size, Focus more on important aspect

↳ Max pooling, Avg pooling

Overfitting

1) Modify the architecture
2) Modify the data set



{ YOLO
Resnet 50

B G P R

B R G P